200 series

SH-SS200

Total Suspended Solids digital sensor

User manual

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Reading instructions

Information for users

Thank you very much for choosing to use our SH-SS200 total suspended solids online sensor (hereinafter referred to as: SS200 sensor). Before using this product, please read this user manual carefully. This manual covers important information and data for the use of the product, which must be strictly observed by the user to ensure the normal operation of the SS200 sensor. The information will help you to use the product correctly and to obtain accurate analysis results.

General description

This manual provides a detailed description of the installation, commissioning, operation and maintenance of the SS200 sensor, as well as describing the measurement principle, instrument composition and performance characteristics of the SS200 sensor.

This manual instructs the user on the proper installation and operation of the SS200 sensor, as well as preventive maintenance work on the SS200 sensor to guarantee the continuous and reliable operation of this sensor.

The products described in this manual have been rigorously inspected before leaving the factory to ensure top quality. In order to ensure safe, high-quality operation and to obtain correct analysis results, the user must strictly follow the instructions described in this manual.

This manual provides all the information required for the proper use of the SS200 sensor. It provides an accurate reference for use by technicians with specialized training or knowledge of the instrument's operational controls. Before operating the instrument, make sure that the safety and warning messages mentioned in this manual are correctly understood and applied to the actual operation.

Delivery and transport

Shipment requirements are in accordance with the terms of the ordering contract.

Please read the appropriate information on the packaging material carefully when opening the box to ensure that the opened goods are complete and undamaged. Please try to retain the outer packaging of the product in case you need to return the meter or parts.

Warranty and maintenance

1) SYSTEA warrants all Products manufactured to be free from defects in material and workmanship under normal use and service. SYSTEA's obligation under any warranty is limited to and Buyer's exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEA's factory, of any defective parts, when returned to SYSTEA by Buyer, transportation prepaid, which SYSTEA's examination discloses to have been factory defective. The time limit of this warranty is 12 (twelve) months from date of shipment of new Products unless otherwise specified by SYSTEA. SYSTEA shall not be liable for consequential damages, in any event. SYSTEA will decide at its only discretion whether to repair or replace a defective part.

For wearing parts and/or in contact with aggressive substances SYSTEA reserves to reduce the warranty period accordingly to the specific application.

2) Repaired Products, all repair work done by SYSTEA is warranted to be free from defects in material or workmanship under normal use and service. SYSTEA's obligation under this warranty is limited to, and the exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEA's factory, of any defective parts installed by SYSTEA, when returned to SYSTEA, transportation prepaid, which SYSTEA's examination discloses to have been factory defective. The time limit of this warranty is three (3) months from date of shipment. SYSTEA shall not be liable for consequential damages, in any event. SYSTEA will decide at its only discretion whether to repair or replace a defective part.

3) THESE WARRANTIES ARE IN LIEU OF ANY WARRANTY OF MERCHANTABILITY, OR FITNESS, OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

4) Expendable Items: this Warranty does not cover expendable or consumable items such as lamps, fuses, parts in contact with fluids, etc.

5) SYSTEA has reasonable period of time for replacements/repairs, and cannot liable for any compensation may arise in relation to the Product

6) SYSTEA only obligation in case of defects, lack of quality or non-conformity of the Products will be repairing or replacing the defective parts. It is agreed that the above-mentioned warranty (i.e.: the obligation to repair or replace the defective parts) is in lieu of any other legal guarantee or liability with the exclusion of any other SYSTEA liability (whether contractual or non-contractual) which may anyhow arise out of or in relation with the Products supplied (e.g. compensation of damages, loss of profit, recall campaigns, etc.).

8) The warranty on goods supplied by SYSTEA is void after 12 months for new Products, if not otherwise specified, 3 months for repairs from delivery date, and in case of one of the following conditions:

a) Due to improper use (water ingress, corrosion, fire, strong electricity strung in, etc.);

b) Damage caused by force majeure (earthquakes, lightning strikes, floods, etc.);

c) Unauthorized alterations within the product without permission;

d) Failure to use the product in accordance with the user's manual and training provisions, causing damage to the product.

e) On products not manufactured by SYSTEA but sold by SYSTEA as results of a distribution contract or as occasional resale, apply the warranty conditions of the manufacturer of the product.

f) Any warranty request, within specified limits, must be notified, the SYSTEA using the form "Warranty Claim" by fax, mail or registered mail, within seven (7) days from the event of the defect. Failing such notification, the Purchaser's right to claim the defects will be forfeited.

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1 Instrument overview

1.1 Instrument Introduction

The total suspended solids online sensor works by RS485 communication interface and standard Modbus protocol and it adopts 90° light scattering method principle. High-precision motor positioning algorithm can achieve the self-cleaning function of the surface; its internal program includes an automatic adjustable gain, which can achieve the accurate measurement of different measurement ranges. The use of corrosion-resistant housing can be applied to different working environments. The device has the advantages of maintenance-free, high precision and simple calibration. The use of ultra-low power components realizes ultra-low power operation of the sensor, which is energy-saving and environmentally friendly.

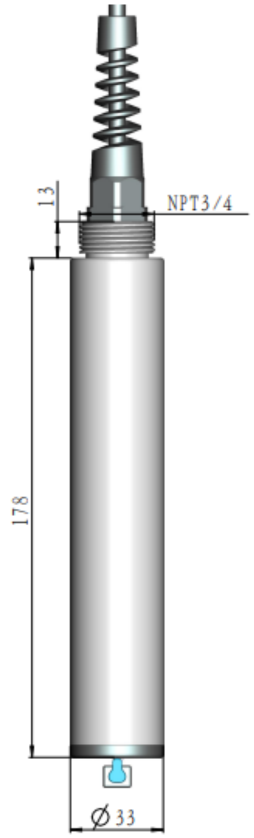

Figure 1.1 Schematic of the appearance of the total suspended solids online sensor

1.2 Working principle

The intensity of the scattered light is measured by passing a beam of light from a stabilized light source, through a cuvette containing the sample to be tested, with the sensor in a position perpendicular to the emitted light. The intensity of the scattered light produced by the beam as it enters the sample is proportional to the concentration of TSS in the sample over a range of concentrations.

1.3 Product features

- Low maintenance
- High stability, low long-term drift, fast response
- Corrosion-resistant housing, waterproof grade IP68, can work underwater for a long time
- RS485 communication interface, standard Modbus protocol for easy integration
- Data analysis software with calibration, recording, analysis and diagnostic functions

1.4 Main uses and scope of application

It is widely used in surface water monitoring, industrial and municipal wastewater monitoring, wastewater treatment, aquaculture, biotechnology, drug development, food and beverage, chemical manufacturing and other industries.

1.5 Environmental conditions of use

Temperature: (0-60) ℃; Pressure: (0-4) bar

2 Technical specification

Working principle	Infrared light scattering
Meas. Wavelength	860nm
Meas. Range	(0~1.000/3.000/10.000) mg/L (other ranges available on request)
Repeatability	±2 %
Accuracy	±5 %
Zero drift (24h)	± 1 %
Range drift (24h)	±5 %
Linearity error	±5 %
Detection limit	1 mg/L
Resolution	0.1 mg/L
Protection class	IP68
Comm. Interface	RS485, standard Modbus protocol, baud rate 9600
Dimensions	(ϕ33*178) mm, cable 10 m (customizable)
Materials	316 Stainless-steel
Mounting threads	Upper NPT3/4 thread
Operating voltage	9~36 Vdc
Power absorption	1 W (motor brushes rotating), 0.4 W (motor brushes not working)
Cleaning method	Mechanical brush
Calibration period	2-3 months
Operating temp.	(0~50) °C

Table 2.1.

3 Use and operation

3.1 installation instructions

The installation of the TSS online sensor has a large impact on its measurement, and the environment in which it is installed and used needs to meet the following requirements:

- When installing the sensor, make sure that the sensor area is not damaged by external collisions, cuts and pressure.
- The minimum water level should be submerged after installation to prevent the electrode from contacting the liquid to be measured when the water level is too low.
- For submerged installation, the sensor must be mounted on the mounting bracket and must not be suspended in water by the sensor cable.
- Submerged installation should be 5cm or more from the bottom and submerged to 2cm or more from the liquid surface.
- The measurement environment needs to be protected from light to prevent the effects of stray light.
- Avoid having external objects blocking the cleaning brush and fiber optic window.

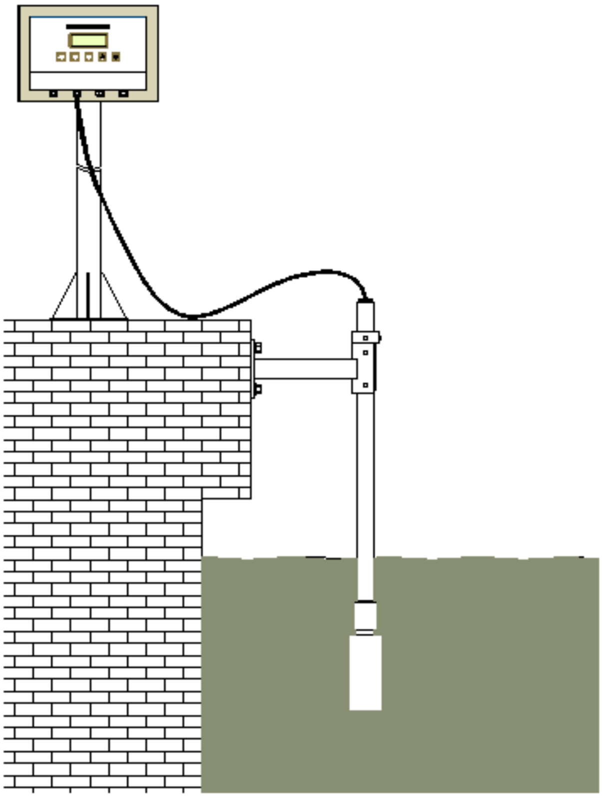

Figure 3.1 Schematic diagram of submerged installation

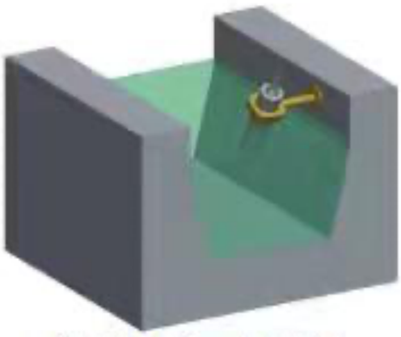

Figure 3.2 Schematic of sensor in channel installation

Figure 3.3 Schematic of sensor in pool installation

Figure 3.4 Schematic of 45° pipe installation

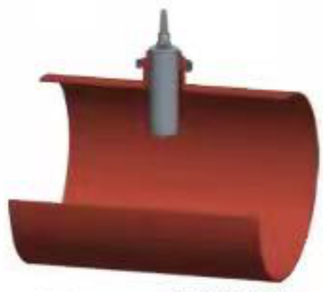

Figure 3.5 Schematic of 90° pipe installation

3.2 Wiring instructions

Sensor type	Wire color	Meaning	Connection type
SS200 total suspended solids online sensor	red	Power positive (+)	Power supply
	black	Power negative (-)	Power supply
	blue	485A	RS-485 communication
	yellow-brown	485B	RS-485 communication

Table 3.1 Wiring description

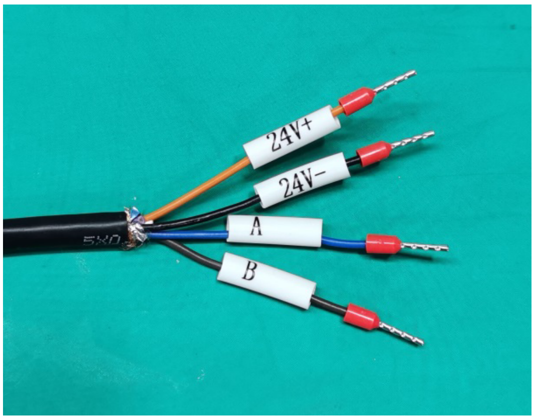

Figure 3.6 Cable wiring

3.3 Operating instructions

After powering up and RS485-communication as described above, the sensor can be connected via the WMC controller / external data-logger or other units. The default baud rate parameter is 9600.

3.4 Communication protocol

baud	9600
data bit	8
stop bit	1
parity check	None
slave address	0x32

Table 3.2 Serial Port Default Parameter Configuration

Modbus-RTU mode, data transfer size end description:

data type	Big and small end model
char (computing)	the main points

uint16	the main points
uint32	Word-ordered little end, byte-ordered big end within word (CDAB)
Float (encoding format IEEE754)	Word-ordered little end, byte-ordered big end within word (CDAB)

Fault code	**Reason for failure**
1ul<<0	Suspended solids over range
1ul<<1	Motor failure
1ul<<2	LED light source alarm
1ul<<4	Hall failure
1ul<<5	Calibration point II anomaly
1ul<<6	Calibration point III anomaly
1ul<<7	Calibration point IV anomaly
1ul<<8	Calibration point V anomaly

Table 3.3 Fault table

Note:

<< is a left-shift operator that first converts the alarm code to binary, with each bit representing an alarm state

For example, 44 is converted to 00000000 00000000 00000000 00101100, which corresponds to fault codes 1ul<<2,1ul<<3,1ul<<5

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Device SN number	ASCII	--	--	12	0x7530	
software version number	ASCII	--		32	0x7536	
Probe Type	uint16		--	2	0x7546	Fixed at 17 for TSS
Probe Status	uint16	--	0-3	2	0x7566	1: Measurement in progress 1: Calibration in progress 2: Motor operation
error code	uint32	--	--	4	0x7567	Uint32_t Big-end format Normal state is 0, see fault table for details
TSS concentration value	Float	mg/L	--	4	0x7569	
Voltage for TSS	Float	V	--	4	0x756b	
Calibration slope of the first segment	Float			4	0x7570	
Calibration of the 1-segment intercept	Float			4	0x7572	
Calibration point 1 voltage	Float			4	0x7574	
Calibration point 2 voltage	Float			4	0x7576	
Calibration slope of the second segment	Float			4	0x7578	
Calibration of 2 intercept	Float			4	0x757a	
Calibration point 3 voltage	Float			4	0x757e	

Demarcation of paragraph 3a	Float			4	0x7580	
Demarcation of paragraph 3b	Float			4	0x7582	
Marked paragraph 3c	Float			4	0x7584	
Calibration point 4 voltage	Float			4	0x7588	
Calibration point 5 voltage	Float			4	0x758a	

Table 3.4 Input register protocols

Functionality	**Read and Write Properties**	**Typology**	**Meas. unit**	**Range of values**	**Byte count**	**Holding Register Address**	**Note**
Device address	Read/Write	uint16	--	0 to 247	2	0x9c40	Default value 0x32
Baud	Read/Write	uint32	--	2400-115200	4	0x9c41	Default value 9600
Over range alarm value	Read/Write	float	--	0-255	4	0x9c45	Default 40000
Motor operating cycle	Read/Write	float	Minutes	--	4	0x9c49	Default 30

Motor compensation value	Read/Write	uint16	--	--	2	0x9c4d	Default 0
Restore Factory Settings	Write	uint16	--	0-1	2	0x9c51	Automatic reset after writing
Motor control	Write	uint16	--	0-1	2	0x9c53	
Restart SW	Write	uint16	--	0-1	2	0x9c54	Automatic reset after writing
Number of motor turns	Write	uint16	--	1-5	2	0x9c57	
Suspended solids calibration time 1	Read/Write	uint32	s	--	4	0x9c60	UNIX timestamp (calibration starts after writing)
Suspended solids calibration value 1	Read/Write	float	--	--	4	0x9c62	
Suspended solids calibration time 2	Read/Write	uint32	s	--	4	0x9c64	UNIX timestamp (calibration starts after writing)
Suspended solids calibration value 2	Read/Write	Float	--	--	4	0x9c66	
Suspended solids calibration time 3	Read/Write	uint32	s	--	4	0x9c68	UNIX timestamp (calibration starts after writing)
Suspended solids calibration value 3	Read/Write	Float	--	--	4	0x9c6a	

Suspended solids calibration time 4	Read/Write	uint32	s	--	4	0x9c6c	UNIX timestamp (calibration starts after writing)
Suspended solids calibration value 4	Read/Write	Float	--	--	4	0x9c6e	
Suspended solids calibration time 5	Read/Write	uint32	s	--	4	0x9c70	UNIX timestamp (calibration starts after writing)
Suspended solids calibration value 5	Read/Write	Float	--	--	4	0x9c72	

Table.3.5 Saving registers

3.5 MODBUS-RTU protocol example (address default 0x32)

3.5.1 Read suspended solids values

Send: 32 04 75 69 00 02 be 18

0x32:Device address 0x04:Read input register function code 0x7569:Input register start address 0x02:Read 2 registers 0xbe18:Check digit

Received: 32 04 04 c0 00 44 66 77 ad

0x32:Device address 0x04:Read input register function code 0x04:4 bytes

0xc0004466: Suspension 0x4466c000(923)

0x77ad:check digit

3.5.2 Suspended solids1 calibration

Send: 32 10 9c 60 00 04 08 05 39 65 0c 00 00 00 00 00 e5 ba

0x32:Device address 0x10:Write multiple registers function code 0x9c60:Write register start address 0x04:Write 4 registers 0x08:Write 8 bytes

0x0539650c:timestamp 0x650c0539(1695286585)2023-09-21 16:56:25

0x00000000:Calibration value 0x00000000(0)

0xe5ba:check digit

Received: 32 10 9c 60 00 04 ea 47

0x32:Device address 0x10:Write multiple registers function code 0x9c60:Write register start address 0x04:Write 4 registers 0xea47:Checksum code

3.5.3 Motor operation

Send: 32 06 9c 53 00 01 93 88

0x32:Device address 0x06:Write single register function code 0x9c53:Write register address 0x0001:Parameter 1 0x9388:Check digit

Received: 32 06 9c 53 00 01 93 88

0x32:Device address 0x06:Write single register function code 0x9c53:Write register address 0x0001:Parameter 1 0x9388:Check digit

3.5.4 Setting the motor run cycle

Send: 32 10 9c 49 00 02 04 00 00 41 f0 0e a3

0x32:Device address 0x10:Write multiple registers function code 0x9c49:Write register start address 0x02:Write 2 registers 0x04:Write 4 bytes

0x000041f0:Calibration value 0x41f00000(30)

0x0ea3:check digit

Received: 32 10 9c 49 00 02 bb 8d

0x32:Device address 0x10:Write multiple registers function code 0x9c49:Write register start address 0x02:Write 2 registers 0xbb8d:Check digit

1 Calibration and maintenance

1.1 Sensor calibration

The sensor is calibrated on three points using a calibration cylinder, as figure below.

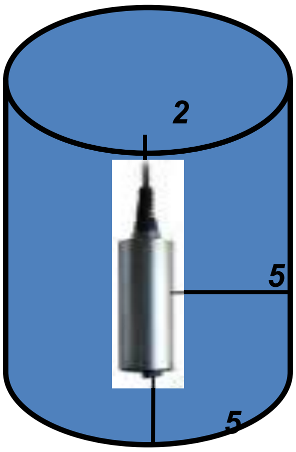

Figure 4.1 Schematic diagram of field water sample calibration

1.1.1 Zero calibration

Pour pure water into the test fixture, put the probe into the pure water sample and input the value of suspended solids concentration 0, click the calibration button to start calibration, wait for the calibration to be completed and then carry out the second point calibration.

1.1.2 Low concentration calibration (50 mg/L)

Shake well the calibration solution 1 well and pour it into the test fixture, put the probe into the sample calibration solution 1 and enter the value of the concentration of suspended solids in the calibration solution 1, click on the calibration button of the calibration solution 1 to start the calibration, and then carry out the third point of calibration after the calibration of the calibration of the calibration solution 1 is finished.

1.1.3 Range calibration (2000 mg/L)

Shake well the calibration solution 2 and pour it into the test fixture, put the probe into the sample calibration solution and input the value of the concentration of suspended solids in the calibration solution 2, click the calibration solution 2 button to start the calibration, and when the calibration is completed, read K\B to see if the calibration information is updated, and end the calibration process after the calibration is completed without any error.

1.2 Sensor maintenance

Generally the sensor should be recalibrated every 2-3 months; if the motor does not turn during operation and other faults should be dealt with in a timely manner, as well as the surface caused by scratches should be replaced in a timely manner. When the light source abnormal alarm needs to be replaced in time.

1.3 Wiper replacement

- Power ON: turn on the sensor. When the wiper brush stops in the middle of the optical window, immediately turn off the power;
- loosen Screws: Use an M1.5 hex key to loosen the fixing screws on both sides of the wiper. Use needle-nose pliers to pull out the wiper that needs to be replaced;
- install new wiper: Insert the new wiper. Use the M1.5 hex key to tighten the fixing screws on both sides.

1.4 Sensor storage

The sensor head needs to be covered with a protective sleeve to prevent abrasion of the brush head as well as at the fiber optic port.